

SOFTWARE PROJECTS MANAGEMENT

Software Engineering Seminar

Author: Eng. Carlos Andrés Sierra, M.Sc.
cavirguezs@udistrital.edu.co

Full-time Adjunct Professor
Computer Engineering Program
School of Engineering
Universidad Distrital Francisco José de Caldas

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Outline

1 Agile Methodologies

2 Project Management



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Agile Methodologies

- Emphasize iterative development, customer collaboration, and flexibility. 2001 \$ ↑ 90's
- They are based on the Agile Manifesto, which values individuals and interactions over processes and tools. 12 items
- Agile methodologies are suitable for projects with rapidly changing requirements and high uncertainty. ~ 2 weeks
- They promote adaptive planning, evolutionary development, and early delivery of valuable software. teams client product

3 → \$5.000.000

1 → \$10.000.000

\$25.000.000 →

150.000.000

marketing?
cloud?

quality



Agile Manifesto Principles

- 1. Customer satisfaction through early and continuous delivery of valuable software. *~ weeks*
- Welcoming changing requirements, even late in development. *business process*
- Delivering working software frequently, with a preference for shorter timescales. *dev ~ max 2 days* *client ~ max 3 weeks*
- Close, daily cooperation between business people and developers. *client* *Product owner*
- Motivated individuals should be trusted to get the job done. *lead*

if it is need

lead



Agile Methodologies Characteristics

- Simplicity is essential, focusing on the essential features.
- Self-organizing teams are encouraged to make decisions.
- Face-to-face communication is preferred for effective collaboration.
- Regular reflection on how to become more effective, and adjustment of behavior accordingly.

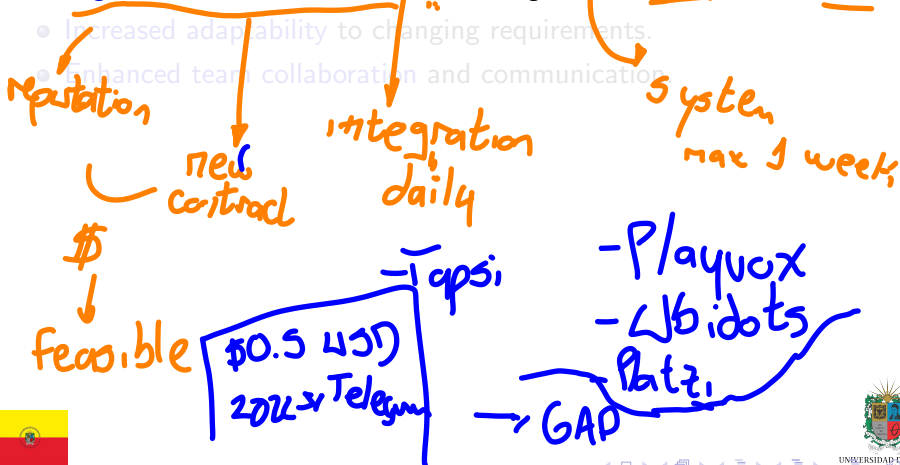
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Agile Methodologies Benefits

- Faster delivery of software features. *~ 1 week*
- Improved quality through continuous testing and feedback.
- Higher customer satisfaction due to regular involvement and feedback.



Agile Methodologies Benefits

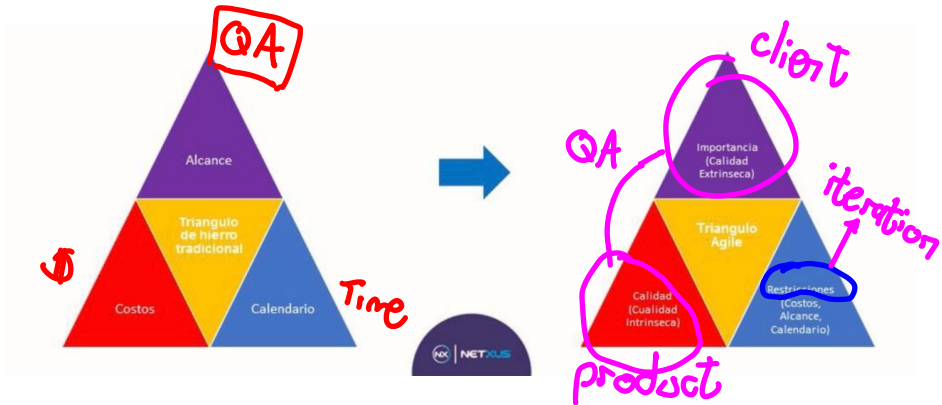
- Faster delivery of software features.
- Improved quality through continuous testing and feedback.
- Higher customer satisfaction due to regular involvement and feedback.
- Increased adaptability to changing requirements.
- Enhanced team collaboration and communication.

daily

through the day

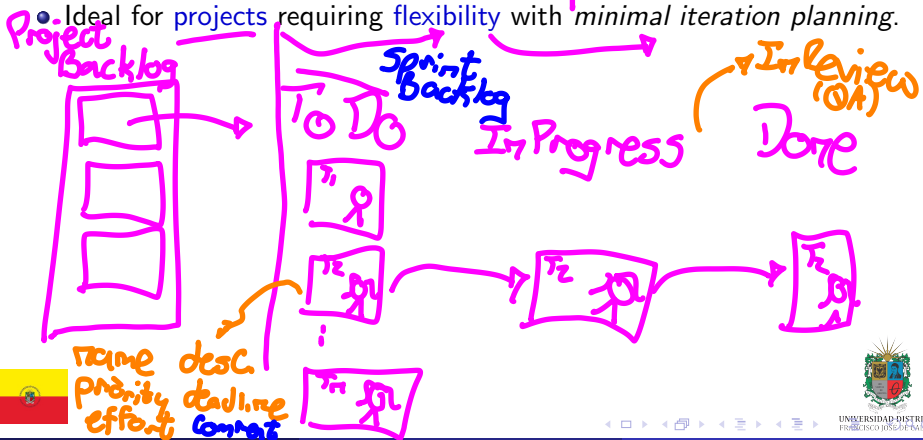


Project Triangles



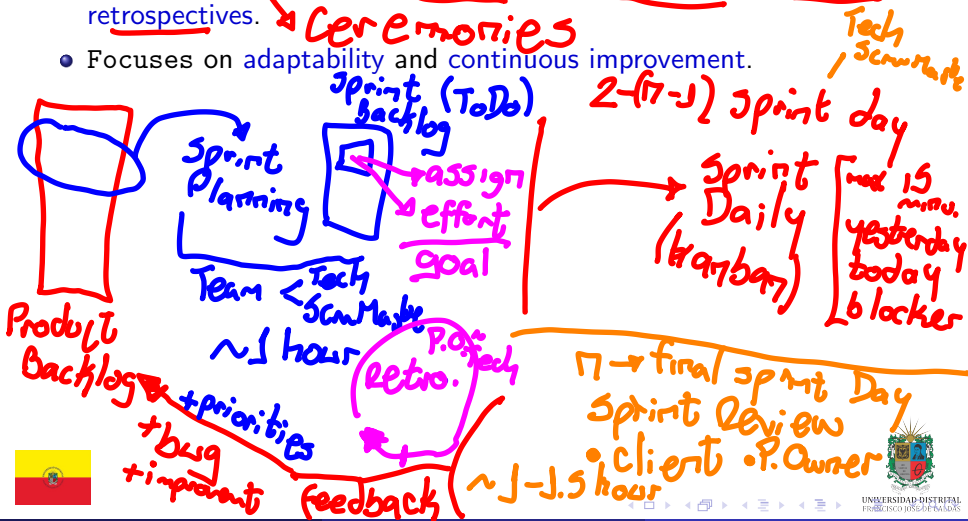
Case Study: Kanban → Tool

- **Kanban** visualizes work items on boards and limits Work In Progress (WIP).
- Emphasizes gradual improvements, flow management, and continuous delivery.
- Ideal for projects requiring flexibility with *minimal iteration planning*.



Case Study: Scrum ~ 1994 → 2005 (6)

- **Scrum** employs short, time-boxed iterations called **sprints**. ^{2 weeks}
- Key practices include daily stand-ups, sprint planning, reviews, and retrospectives.
- Focuses on adaptability and continuous improvement.



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Enterprises: Bottom-Up and Top-Down Approaches

- **Bottom-Up Approach:** Analyzes an enterprise by examining its individual units or components, then aggregating them to *understand* the entire organization.
- **Top-Down Approach:** Starts with an overall vision or strategy and decomposes it into subsystems, departments, and processes.



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PIECE Framework for Enterprises

- **Participation:** *Engaging stakeholders* at every level.
- **Independence of Thought:** Encouraging *diverse, innovative ideas*.
- **Elaboration:** Developing and *refining ideas* and *processes*.
- **Communication:** Ensuring clear, *effective exchange* of information.
- **Exploration:** Embracing *continuous innovation* and *improvement*.



Enterprise System Typologies

- **Rational Systems:** Organizations driven by logical, structured processes and clear hierarchies.
- **Natural Systems:** Organizations viewed as self-organizing entities with emergent behavior.
- **Open Systems:** Enterprises that continuously interact with their external environment for information, resources, and innovation.



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Business Systems and Models

- **Business Systems:** Frameworks that encompass an enterprise's internal processes, operations, and strategies.
- **Examples:** ERP systems, CRM systems, SCM systems.
- **Business Models:** Describe how an organization creates, delivers, and captures value.
 - Examples include subscription-based, freemium, platform-based, and direct sales models.



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Business Models Examples

Understanding Business Models Through Flywheels

amazon



NETFLIX



Spotify



LinkedIn



Project Management in Software Engineering

- **Project management** is the process of **planning**, **executing**, and **controlling** software projects to achieve specific **goals**.
- It involves defining **project scope** (*objectives, requirements, boundaries, ...*), **allocating resources** (*human, financial, technical, ...*), **scheduling tasks** (*time estimation, task dependencies, ...*), **managing risks** (*identifying, assessing, mitigating, ...*), **managing changes** (*change requests, impact analysis, ...*), **monitoring progress** (*tracking milestones, deliverables, ...*), and **ensuring quality**.
- **Effective project management** is crucial for **delivering software** projects on time, within budget, and meeting customer expectations.



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Project Management Key Points

- Choosing the right **methodology** for your project is crucial. Consider factors like: project size, complexity, team experience, and customer requirements.
- Build a **strong team** with diverse skills and expertise. Encourage *collaboration, communication, and knowledge sharing*.
- Define clear **goals** and **objectives** for your **project**. Ensure that all team members understand the project vision and their roles in achieving it.
- Use **tools** and **techniques** to support project management, such as project management software, version control systems, issue tracking systems, and collaboration tools.
- Regularly **review** and **adjust** your project plan based on feedback and changing circumstances to ensure *continued alignment* with project goals.



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Thanks!

Questions?



Repo: www.github.com/EngAndres/ud-public/tree/main/courses/software_engineering_seminar

